CHAPTER – 1

INTRODUCTION

The rapid advancement of intelligent transportation systems and autonomous driving technologies has created a strong need for reliable communication between vehicles. Vehicle-to-Vehicle (V2V) communication is a wireless communication method that enables vehicles to share essential information such as speed, position, acceleration, braking status, and surrounding hazards in real time. By allowing vehicles to "talk" to each other, V2V communication significantly enhances a vehicle's awareness of its environment beyond the limitations of cameras, sensors, or human vision alone.

V2V communication is becoming a fundamental component of autonomous driving. Self-driving vehicles rely heavily on accurate, real-time data to make safe and informed decisions. Through direct communication with nearby vehicles, an autonomous car can anticipate potential collisions, perform coordinated lane changes, adapt to dynamic traffic conditions, and respond to sudden obstacles more efficiently. This level of cooperative behavior improves road safety, reduces reaction time during emergencies, and supports smoother traffic flow, ultimately contributing to safer and smarter transportation systems.

Testing V2V systems directly in real-world traffic is expensive, risky, and difficult to control. Therefore, simulation plays a critical role in developing and validating these technologies. Simulation environments allow developers to test communication protocols, encryption mechanisms, decision-making algorithms, and multi-agent behavior under controlled, repeatable conditions. Scenarios such as high-speed traffic, complex intersections, unpredictable obstacles, and emergency maneuvers can be reproduced safely without endangering people or property.

In this project, the MetaDrive Simulator is used to create a realistic multi-agent driving environment where vehicles interact with each other through secure V2V communication. By integrating networking, encryption, and autonomous control within the simulation, this project demonstrates how V2V technology can improve cooperation between vehicles and enhance decision-making in autonomous driving systems.

1.1 Overview

The evolution of autonomous and connected vehicles has transformed modern transportation systems, making them smarter, safer, and more efficient. A key technology driving this transformation is Vehicle-to-Vehicle (V2V) communication, which enables

vehicles to exchange critical information with surrounding vehicles in real time. This exchange includes data such as vehicle position, speed, acceleration, braking intention, lane changes, and detection of nearby obstacles. Such direct communication allows vehicles to anticipate threats earlier, coordinate movements, and collaboratively navigate complex traffic environments.

V2V communication acts as an additional layer of intelligence that complements traditional sensors like LiDAR, radar, and cameras. While sensors provide visual or physical awareness, V2V communication delivers predictive awareness, allowing vehicles to "see" beyond their immediate field of view. This capability is essential for preventing collisions, reducing traffic congestion, and ensuring smoother vehicular interactions—especially in crowded urban areas, intersections, and high-speed scenarios. Therefore, V2V communication forms the backbone of future intelligent transportation systems (ITS).

Developing and validating V2V communication in the real world is challenging due to safety concerns, unpredictable conditions, and high costs associated with physical testing. To overcome these limitations, simulation-based development has become the preferred approach in the research and automotive industries. Simulators provide a controlled, repeatable, and safe environment where different communication protocols, sensor models, encryption mechanisms, and decision-making algorithms can be designed and evaluated without real-world risks.

In this project, a secure and scalable V2V communication framework is implemented using the MetaDrive Simulator, an advanced simulation platform designed for reinforcement learning and autonomous driving research. MetaDrive supports highly realistic driving environments, multi-agent interactions, and customizable vehicle behaviors, making it ideal for testing communication and decision-making systems.

The core components of the project include:

Multi-Agent Vehicle Simulation:

Multiple vehicles are spawned in the MetaDrive environment, each operating as an independent intelligent agent capable of movement, perception, and decision-making.

Real-Time Communication Engine:

A custom communication layer based on UDP socket programming enables vehicles to broadcast their state data—such as coordinates, speed, and detected obstacles—to all other vehicles within the network.

Secure Data Transmission using AES Encryption:

To ensure confidentiality and prevent unauthorized access, all communication packets are encrypted using the Advanced Encryption Standard (AES). This protects the integrity of the data exchanged between vehicles and models real-world cybersecurity requirements for intelligent vehicles.

Receiver Module for Data Decoding & Processing:

Each vehicle includes a receiver component responsible for listening to incoming packets, decrypting the data, validating message structure, and extracting useful information for decision-making.

Integration with Vehicle Control Logic:

Received V2V data can be used by the vehicle to adjust its movement, avoid collisions, respond to obstacles, and coordinate with other agents—forming the basis for cooperative autonomous driving.

This integrated system demonstrates how secure V2V communication enhances vehicle awareness and supports collaborative navigation within a multi-agent simulation. It highlights the importance of communication security, low-latency message transmission, and reliable system modeling in the development of next-generation autonomous vehicles.

Through this project, a complete pipeline—combining simulation, communication, and cryptography—is developed to analyze and validate the real-time behavior of connected autonomous vehicles. The framework can be extended further for real-world datasets, reinforcement learning, V2I (Vehicle-to-Infrastructure) communication, platoon formation, and safety-critical decision modeling.

1.2 Problem Statement

- **Limitations of Traditional Onboard Sensors:** Cameras, LiDAR, and radar suffer from line-of-sight, range, and environmental constraints, leading to delayed hazard detection and reduced situational awareness.

- **Challenges in Real-World V2V System Development:** Implementing and testing V2V communication on actual vehicles is difficult due to high costs, safety risks, and unpredictable real-world conditions.

- **Security Concerns in V2V Communication:** V2V systems face vulnerabilities such as data tampering, malicious attacks, and lack of proper encryption, potentially leading to dangerous vehicle behavior.
- **Need for Reliable Real-Time Communication:** Issues like latency, loss of integrity, and synchronization failures in multi-vehicle communication affect coordination and overall road safety.
- **Absence of an Integrated Simulation Framework:** There is a lack of controlled platforms that can simulate multi-vehicle environments, test encrypted V2V communication, and analyze its impact on autonomous decision-making.
- **Requirement for Safe Evaluation Environment:** A secure simulation setup is needed to study interactions, communication reliability, and safety outcomes without exposing real vehicles to risks.

1.3 Significance and Relevance of Work

- **Enhances Situational Awareness and Traffic Safety:** V2V communication extends a vehicle's perception beyond onboard sensor limitations, improving collision avoidance, lane coordination, and overall traffic efficiency.
- **Ensures Secure and Reliable Data Exchange:** The use of AES encryption protects V2V messages from tampering and unauthorized access, addressing critical cybersecurity concerns without compromising real-time performance.
- **Enables Safe and Scalable Multi-Agent Testing:** Using the MetaDrive Simulator allows controlled, repeatable, and realistic testing of autonomous vehicle interactions, communication protocols, and encryption schemes—offering a safe platform for validating cooperative driving behaviors.

1.4 Objectives

Primary Objectives

- To design and implement a secure Vehicle-to-Vehicle (V2V) communication system using Python and networking protocols.
- To simulate real-time multi-agent vehicle interactions using the MetaDrive autonomous driving simulator.
- To integrate AES-based encryption to ensure secure, tamper-proof data exchange between vehicles.

Secondary Objectives

- To develop a broadcaster–receiver communication architecture using UDP for low-latency data transmission.
- To enable vehicles to exchange key information such as position, speed, timestamp, and obstacle data.
- To model and test cooperative driving behavior supported by shared communication data.
- To analyze how secure V2V communication affects situational awareness and safety in autonomous driving. To design and implement an IoT-based greenhouse monitoring and control system.
- To automate irrigation, lighting, and ventilation using intelligent decision logic.
- To incorporate AI-based plant disease detection using CNN classification.
- To analyse environmental trends and dependencies that influence plant growth.
- To build a scalable intelligent greenhouse system with future expansion capability.

1.5 Methodology

The methodology adopted for this project involves the systematic design, development, and integration of secure Vehicle-to-Vehicle (V2V) communication within a simulated multi-agent driving environment. The overall workflow follows a structured pipeline beginning from environment setup, communication architecture design, encryption integration, simulation execution, and final performance evaluation. The major steps involved are described below:

• **Requirement Analysis:** Identifying the functional needs of secure V2V communication, including broadcasting, receiving, real-time processing, and the necessity of encryption for confidentiality and message integrity. Selecting MetaDrive as the simulation platform for multi-agent autonomous driving.

• **System Design:** Creating a modular architecture separating simulation logic, communication modules, encryption handlers, and configurations. Defining communication protocols, data formats, and an overall structure including the Simulation Layer, Communication Layer, Encryption Layer, and Decision Logic Layer.

• **MetaDrive Simulation Setup**: Installing and configuring MetaDrive for multi-agent environments, setting parameters such as number of vehicles, scenarios, and observation space. Implementing env_manager.py to initialize MultiAgentMetaDrive

and enabling real-time rendering.

- **Communication Module Development:** Broadcaster: Implementing a UDP-based broadcaster to collect vehicle state data, serialize it using JSON, and send periodic messages.
- **Receiver:** Creating a UDP receiver to listen for incoming packets, deserialize them, validate structure, and prepare the data for decision-making.
- **Encryption Integration:** Implementing AES-based encryption using an Encryption Manager configured through v2v_settings.yaml. Encrypting outgoing messages with AES-CBC and decrypting incoming data while ensuring minimal latency impact.
- **Multi-Agent V2V Integration:** Connecting broadcaster and receiver modules to each vehicle so each agent can send and receive data. Synchronizing communication frequency with simulation steps and maintaining ordering using timestamps when needed.
- **Simulation Execution:** Running the complete system using run_all.py, launching the environment and communication modules as separate processes. Monitoring logs for message broadcasting, encryption accuracy, decryption success, and simulation performance.

1.6 Organization of the Report

- **Chapter 1:** Introduction, this chapter provides an overview of the project, introduces the concept of Vehicle-to-Vehicle (V2V) communication, and highlights the challenges faced in autonomous driving systems related to safety, coordination, and secure data exchange. It also outlines the motivation, objectives, scope, significance, and relevance of the work.
- **Chapter 2:** Literature Survey, this chapter reviews previous research in the fields of V2V communication, intelligent transportation systems, autonomous driving simulations, and secure communication protocols. It examines existing studies on multi-agent driving environments, MetaDrive-based simulation frameworks, and the application of cryptographic techniques such as AES in vehicular networks.
- **Chapter 3:** System Requirements Specification, this chapter describes the functional requirements (simulation setup, broadcasting, receiving, encryption, communication flow) and non-functional requirements (performance, reliability, security, scalability, and real-time constraints). software requirements, and external dependencies such as Python libraries and MetaDrive Simulator.

- **Chapter 4:** Analysis and Design, this chapter focuses on the architectural aspects of the system. It includes the system architecture diagram, block diagram, data flow diagram, and workflow of the V2V communication framework. It explains the design of the communication layer (broadcaster and receiver), encryption module, configuration files, and integration with the MetaDrive multi-agent environment.
- **Chapter 5:** Implementation, this chapter explains the development of each module including the MetaDrive environment setup, vehicle initialization, broadcaster module, receiver module, AES encryption manager, and integration of all components through the master script (run_all.py). It also details the communication pipeline, real-time message exchange, and simulated vehicle interactions.
- **Chapter 6:** Results and Discussion, this chapter presents the output of the simulation, including logs, screenshots, encrypted message samples, and V2V communication behavior. It analyzes system performance, encryption impact, multi-vehicle coordination, and overall effectiveness of the implemented framework.
- **Chapter 7:** Conclusion and Future Enhancements, this chapter summarizes the work, highlights key findings, and discusses how secure V2V communication improves autonomous vehicle safety and coordination. It also suggests future enhancements such as integrating V2I (Vehicle-to-Infrastructure), reinforcement learning, advanced decision-making algorithms, or real-world vehicular datasets.
- **Bibliography:** his section includes research papers throughout the project.

CHAPTER – 2

LITERATURE SURVEY

2.1 Overview

Vehicle-to-Vehicle (V2V) communication, autonomous driving, and multi-agent simulation systems have become major research areas in intelligent transportation technologies. With increasing road traffic density and the advancement of autonomous vehicles, researchers have focused on developing communication mechanisms that enable vehicles to share real-time information to improve safety, efficiency, and coordination. Literature in this domain extensively explores vehicular communication protocols, wireless networking models, and cooperative awareness systems that allow vehicles to anticipate and respond to environmental conditions more effectively.

At the same time, simulation platforms have gained significant importance due to the complexity and risks associated with real-world testing. Researchers have developed a variety of simulators—such as CARLA, SUMO, and MetaDrive—to evaluate autonomous vehicle behavior in controlled, repeatable, and safe environments. These simulators support multi-agent interactions, reinforcement learning, and customizable traffic scenarios, making them well-suited for studying V2V communication and cooperative driving systems.

Another major area of research focuses on securing vehicular communication. As V2V systems rely on wireless data exchange, they are vulnerable to attacks such as spoofing, tampering, and message replay. Thus, encryption methods such as AES, PKI, and digital signatures have been widely studied to ensure confidentiality, authenticity, and integrity of transmitted messages, especially in safety-critical applications like autonomous driving.

The existing literature highlights both the potential and challenges of integrating real-time communication, autonomous driving behavior, and cybersecurity. This project builds upon these research findings by implementing a secure V2V communication framework within the MetaDrive simulator, addressing gaps in combined simulation-based communication security and multi-agent interaction.

Paper 1: "Deep Learning for Safe Autonomous Driving: Current Challenges and Future Directions"

Authors: Wei Xu, Fei-Yue Wang et al.

This paper presents a comprehensive survey of how **Deep Learning (DL)** is transforming safe autonomous driving by improving perception, decision-making, and control systems. The authors highlight that autonomous driving heavily depends on accurate sensing and reliable interpretation of road environments, including lane detection, vehicle and pedestrian recognition, drowsiness monitoring, collision avoidance, and traffic sign understanding. The paper explains that recent advancements in DL—especially convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid learning architectures—have significantly enhanced the performance of these tasks, outperforming traditional handcrafted feature methods.

The authors emphasize that although DL models show remarkable accuracy, **real-world deployment still faces challenges** such as unpredictability, lack of extensive training data, hardware limitations, and difficulties in handling rare or extreme driving conditions. They also point out that safety-critical tasks require extremely robust and interpretable models, but most DL models operate as "black boxes," creating concerns about transparency and trust. The paper further warns that dependency on multiple system components (sensors, cameras, algorithms) means failure in even one module can compromise safety.

To address these issues, the survey outlines current research directions, including **multi-sensor fusion**, **self-supervised learning**, **uncertainty estimation**, and **explainable AI (XAI)** for improving reliability. The paper also reviews evaluation metrics used for benchmarking DL models in autonomous driving. Overall, the study identifies existing gaps in DL-based autonomous driving systems and recommends future research to enhance robustness, interpretability, scalability, and safe real-time performance.

Paper 2: A Survey on Datasets for the Decision Making of Autonomous Vehicles

Authors: Yuning Wang, Jianqiang Wang et al.

This survey paper provides a comprehensive review of datasets specifically used for **decision-making in autonomous vehicles (AVs)**—a domain that has received less attention than perception datasets. The authors highlight that while perception datasets (object detection, tracking, segmentation) are well-studied, there is a lack of structured

information on datasets designed to support **behavior planning, trajectory generation, and driver intelligence modeling**. The paper begins by explaining that decision making is a crucial module in autonomous driving architectures, acting as the "brain" that links perception to control. As driving scenarios grow more complex, **data-driven methods (machine learning, deep learning, reinforcement learning)** increasingly outperform rule-based approaches, but their effectiveness depends heavily on high-quality datasets.

The authors categorize decision-making datasets into **three major sources**:

1. **Vehicle-related data** – status (GPS/IMU), manipulation data (steering, throttle, brake), and internal CAN bus signals.

2. **Environment-related data** – road semantics, interactions, traffic participants, HD maps, weather, anomalies.

3. **Driver-related data** – gaze, attention, posture, behaviors, and physiological states. This classification helps researchers select datasets aligned with their research needs.

The survey also traces the **historical evolution** of AV decision datasets, noting a massive increase in size, diversity, and complexity—from early datasets like NGSIM to modern ones such as Waymo Open Motion, unplan, Lyft L5, and INTERACTION. These new datasets emphasize **challenging scenarios**, richer annotations, and multi-sensor setups (LiDAR, radar, semantic maps), enabling realistic modelling of lane changes, interactions, and accidents.

The paper further explores dataset applications across three decision-making routes:

- **Environment Understanding** (context analysis, intention prediction, trajectory prediction)

- **Driver Intelligence Learning** (attention tracking, behavior analysis)

- **Decision Learning** (behavior strategy, trajectory planning, manipulation learning) Each category is supported with examples of widely used datasets such as highD, JAAD, PIE, Argoverse, Brain4Cars, HDD, KITTI, and NGSIM.

Key limitations identified include:

- Lack of unified data standards (different formats like JSON, TXT, ROS bag).

- Limited driver-focused datasets (attention, posture, physiological signals are fragmented).
- Insufficient datasets containing **all three data types** (driver + vehicle + environment) synchronized together.
- Difficulty in using datasets interchangeably due to heterogeneous annotation styles.

Overall, the survey emphasizes that **future AV decision-making research requires richer, more integrated, and standardized datasets** that support multi-modality, closed-loop simulation, and challenging driving conditions. This work provides an essential guide for researchers choosing appropriate datasets for decision-making models, trajectory planning, and safe autonomous driving.

Paper 3: "Challenges and Solutions for Cellular-Based V2X Communications"

Authors: Arshad Raza, Haitham Cruickshank et al.

This paper presents a thorough analysis of **Cellular Vehicle-to-Everything (C-V2X)** communication, focusing on the technological challenges and potential solutions needed to support next-generation connected and autonomous vehicles. The authors highlight that C-V2X, built on LTE and evolving into 5G New Radio (NR), aims to provide reliable low-latency communication for safety-critical applications such as collision avoidance, cooperative driving, platooning, and remote vehicle control. However, despite its advantages over traditional DSRC, several limitations hinder its large-scale deployment.

A central challenge discussed is **latency**, especially for ultra-reliable low-latency communication (URLLC). Safety applications require end-to-end delays below 10 ms, but cellular networks often struggle with scheduling delays, congestion, and handover issues. Interference management is another critical problem due to dense vehicular environments, high mobility, Doppler shifts, and the need to support thousands of simultaneous V2X links. The paper also explains issues related to **resource allocation**, where dynamic and decentralized scheduling is essential to prevent packet collisions in sidelink Mode 4 communication.

The authors examine **network coverage gaps**, especially in rural and highway environments where cell towers are sparse, affecting reliability. They also discuss **mobility management challenges**, as vehicles moving at high speeds must frequently switch

between cells, potentially causing communication dropouts. Scalability becomes problematic in urban regions with a high density of vehicles transmitting frequent safety messages.

To address these challenges, the paper proposes solutions such as **advanced scheduling schemes**, adaptive modulation, multi-access edge computing (MEC) to reduce latency, and beamforming to improve link reliability. For interference and congestion problems, the authors recommend hybrid resource allocation, cooperative sensing, and machine-learning-based traffic prediction. 5G NR-V2X is highlighted as a major improvement due to flexible numerologies, slot-based scheduling, improved sidelink reliability, and support for enhanced V2X use cases like cooperative perception.

Security is also discussed as a major concern. Since V2X communications involve broadcasting safety messages, the system is vulnerable to spoofing, jamming, replay attacks, and false information injection. The paper highlights the importance of authentication frameworks, public key infrastructure (PKI), and lightweight encryption mechanisms to ensure data integrity without increasing latency.

Overall, the paper concludes that while C-V2X provides a strong foundation for connected and autonomous vehicles, achieving **high reliability, low latency, scalability, and security** requires continued improvements in 5G and future 6G networks. The study provides a roadmap for enhancing network architecture, resource allocation, and security mechanisms to support real-world deployment of V2X systems.

Paper 4: Longitudinal Control of Automated Vehicles – Integrating Deep Reinforcement Learning with IDM

Authors: Jiajun Wang, Xiaobo Liu et al.

This paper proposes a hybrid longitudinal control approach that combines **Deep Reinforcement Learning (DRL)** with the well-established **Intelligent Driver Model (IDM)** to improve the safety, stability, and comfort of automated vehicle motion planning. Traditional IDM models provide smooth and interpretable car-following behaviour but struggle in complex, dynamic environments where quick adaptation is required. On the other hand, DRL models can learn optimal actions from experience but often suffer from instability, slow convergence, or unsafe decisions during exploration. To address these limitations, the authors integrate both methods into a unified framework.

The proposed system uses **Twin Delayed Deep Deterministic Policy Gradient (TD3)**, a state-of-the-art reinforcement learning algorithm known for improving overestimation bias and stabilizing continuous control learning. The controller learns optimal acceleration and braking commands while IDM contributes structured behaviour and safety rules. According to the block diagrams and system architecture (pages 3–4), the hybrid model takes into account vehicle speed, distance to the leading vehicle, relative velocity, and safe time headway to generate smooth and collision-free longitudinal control.

Simulation experiments demonstrate that the DRL–IDM controller responds more efficiently to sudden braking events, maintains safer following distances, and minimizes oscillations compared to standalone IDM or pure DRL approaches. The graphs on pages 6–8 show improved stability in speed tracking and reduced jerk values, contributing to ride comfort. The hybrid system also generalizes better across varied traffic densities and unpredictable leader behaviours.

One of the key findings is that IDM-based shaping of the reward function accelerates DRL training and prevents dangerous exploration, making the model safer for real-world deployment. The paper also highlights the importance of balancing performance and safety when training policy networks for automated driving. Limitations noted include high computational demand for DRL training and the need for more comprehensive real-world datasets to improve generalization.

In conclusion, this work demonstrates that integrating classical car-following models with modern DRL techniques can significantly enhance longitudinal vehicle control, paving the way for more robust and human-like automated driving behaviour.

Paper 5: Predictive Trajectory Planning Using a Spatiotemporal Risk Field

Authors: Yifan Zhao and Masayoshi Tomizuka *et al.*

This paper introduces a novel **predictive trajectory planning framework** for autonomous vehicles using a **Spatiotemporal Risk Field (STRF)**, which models the risk associated with dynamic obstacles, road structure, and future environment changes. The authors argue that conventional trajectory planners often fail in highly dynamic environments because they rely on instantaneous or short-term predictions, ignoring how risk evolves over time. To overcome this limitation, the paper constructs a 3D risk representation (2D space + time), enabling the vehicle to plan more robust and anticipatory trajectories.

The STRF integrates multiple risk sources, such as obstacle motion, collision probability, time-to-collision (TTC), road boundaries, and kinematic feasibility. As seen in the diagrams and mathematical formulations on pages 4–6, the risk field models not only the positions of surrounding vehicles but also their predicted trajectories over a future horizon. This gives the planning module the ability to foresee risky zones and select safer paths proactively.

The trajectory planner uses a **sampling-based search strategy**, where candidate trajectories are generated and evaluated against the STRF. Each trajectory is assigned a cumulative risk cost based on its interaction with the risk field. The planner then selects the optimal trajectory that minimizes risk while satisfying dynamic constraints such as acceleration, jerk limits, and lane boundaries. Simulation results on pages 8–10 demonstrate that this risk-aware approach significantly improves path smoothness, collision avoidance, and responsiveness in traffic scenarios.

One of the key strengths of this method is its ability to handle **highly dynamic traffic**, such as sudden braking of leading vehicles, lane-changing manoeuvres of adjacent cars, and interactions at intersections. The STRF continuously updates with new perception data, making the planner highly adaptive. The paper also compares the proposed method with classical techniques like potential fields and MPC, showing that STRF leads to fewer collisions and smoother trajectories.

However, the authors note challenges such as computational complexity in dense traffic, the need for accurate motion prediction models, and real-time optimization. Despite these limitations, the STRF-based trajectory planning framework demonstrates excellent potential for enhancing autonomous driving safety and foresight in complex environments.

Paper 6: Real-Time Detection of Malicious Nodes in VANETs using Poplar Optimized Sparsity-Aware Network

Authors: R. Ezumalai, D. Santhakumar

This paper presents a novel real-time framework for detecting malicious nodes in **Vehicular Ad Hoc Networks (VANETs)** using a **Poplar Optimized Sparsity-Aware Neural Network (POSAN)**. VANETs rely on continuous data exchange between vehicles, but their open wireless nature makes them vulnerable to attacks such as false data injection, Sybil attacks, blackhole attacks, and message tampering. The authors highlight that traditional security methods—cryptography, authentication, and trust management—are

not always sufficient for detecting dynamic and intelligent malicious behaviour in real time. Therefore, intelligent intrusion detection mechanisms are needed.

The proposed POSAN model enhances VANET security by analysing vehicle-generated data and identifying abnormal communication patterns. The architecture, as shown in the diagrams on pages 4–6, reduces computational complexity using sparsity-aware layers, enabling faster processing suitable for real-time vehicular environments. The Poplar optimization further improves model pruning, memory efficiency, and inference speed, making it deployable on edge devices inside vehicles.

The paper demonstrates that POSAN effectively classifies nodes as benign or malicious using features like packet delivery ratio, transmission frequency, location mismatch, signal strength variations, and message consistency. The authors validate the framework using real VANET datasets and simulation-based attack scenarios. Experimental results (pages 7–10) show that POSAN outperforms conventional deep learning models such as CNN, LSTM, and MLP in terms of accuracy, detection rate, and false positives. It achieves high precision while maintaining low computational demand, which is critical for vehicular applications where decisions must be made within milliseconds.

One of the significant contributions of the paper is its ability to handle **dynamic adversarial behaviour**. Malicious nodes often mimic normal traffic patterns to evade detection, but POSAN's sparsity-aware learning enhances robustness to such evasion tactics. The model also updates itself incrementally, enabling continuous learning with minimal overhead.

However, the authors acknowledge limitations such as dependency on high-quality labelled datasets, potential performance drops during large-scale network congestion, and challenges in integration with cryptographic frameworks. Despite these limitations, the proposed methodology provides an efficient solution for improving VANET security and ensuring trusted V2V and V2X communication.

2.2 Problem Definition

Autonomous vehicles must communicate with each other to avoid collisions and make safe driving decisions. But testing real V2V communication on roads is expensive, unsafe, and difficult. There is also a risk of attackers sending false messages because V2V data is transmitted wirelessly.

Therefore, the problem is to **create a simulated environment** where multiple vehicles can:

- Communicate with each other in real time,
- Send and receive their position, speed, and surrounding information,
- Use encrypted messaging to ensure secure and trustworthy communication,
- And test how V2V communication improves safety and decision-making.

The goal is to build a **secure V2V communication system** inside the MetaDrive simulator to study safety, coordination, and collision avoidance without real-world risk.

CHAPTER – 3

SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification outlines the functional and non-functional requirements necessary to design, implement, and execute a secure Vehicle-to-Vehicle (V2V) communication simulation using the MetaDrive environment. This chapter defines the system behavior, interface requirements, hardware/software dependencies, and performance expectations.

3.1 System Requirements Specification

The System Requirements Specification (SRS) defines the essential functional and non-functional requirements necessary to design and implement the secure V2V communication system within the MetaDrive simulation environment. It outlines how the system should behave, what features it must provide, and the constraints under which it must operate.

Purpose: The purpose of this project is to design and implement a secure Vehicle-to-Vehicle (V2V) communication system within the MetaDrive simulation environment, allowing multiple autonomous vehicles to exchange real-time information such as speed, position, and surrounding obstacles. The system aims to demonstrate how V2V communication improves road safety, enhances situational awareness, and supports collision-free navigation.

This project also focuses on ensuring **secure message transmission** using AES-based encryption so that the shared information cannot be tampered with or intercepted by malicious entities. By integrating simulation, communication, and security, the project provides a safe and scalable platform to test autonomous vehicle behavior without real-world risks.

3.2 Specific Requirements

The system must support real-time, secure V2V communication between multiple autonomous vehicles inside the MetaDrive simulation. Each vehicle should broadcast and receive essential driving information—such as position, speed, and surrounding obstacles—using low-latency UDP messaging. All messages must be encrypted using AES to ensure data confidentiality and prevent tampering. The simulation should run smoothly

in multi-agent mode, with each vehicle processing received V2V data to improve awareness and avoid collisions. Configurations like vehicle IDs, ports, broadcast intervals, and encryption settings must be easily adjustable through YAML files. Overall, the system should operate reliably, maintain real-time performance, and provide accurate communication for safe autonomous driving behavior.

3.2.1 Hardware Specification

- **Recommended**, these specifications support advanced simulations, higher FPS rendering, large-scale multi-agent traffic, encryption-heavy workloads, and future ML-based modules.

- **Processor** (High Performance CPU), Intel Core i7/i9 (10th Gen or above), AMD Ryzen 7 / Ryzen 9 (3000 series or above), Recommended Cores: 8–16 cores
  Reason: Multi-processing for broadcaster, receiver, and environment modules benefits from multi-core design.
- **Memory** (High-Capacity RAM), 32 GB – 64 GB DDR4/DDR5
  Reason: Multi-agent simulation + real-time rendering + encryption + logging demands large memory headroom.
- **Graphics Card** (High-End GPU), Recommended: NVIDIA RTX 3060 / 4060
  Reason: MetaDrive uses Panda3D rendering; smoother FPS at high settings and larger maps requires strong GPU capacity.
- **Storage (Fast IO Performance)**, 512 GB – 1 TB NVMe SSD
  Reason: Faster dependency loading, large dataset handling, faster startup, and smooth caching.

3.2.2 Software Specification

- **Operating System:** Windows 10/11 or Ubuntu

- **Coding Languages:** Python 3.10.x

- **Packages & Libraries:** MetaDrive 0.4.3, PyYAML, NumPy, Pandas, SciPy, Cryptography / PyCryptodome, Socket programming (default Python module), Matplotlib, Seaborn, Environment Management: Virtual environment (venv) .

- **Code Editor:** VS Code / PyCharm

3.3 Functional Requirements

- **MetaDrive initialization,** Start MetaDrive in multi-agent mode with configurable map, seed, and render options.
- **Vehicle management,** Create and maintain multiple vehicle agents (IDs, pose, speed, sensors). Update each vehicle's state every simulation step.
- **Broadcasting,** each vehicle must periodically broadcast its state (position, speed, heading, obstacles, timestamp), Broadcasting uses UDP and configurable interval.
- **Receiving,** each vehicle must listen on a dedicated UDP port for incoming V2V messages, received messages must be parsed, validated and queued for processing.
- **Encryption,** optionally encrypt outgoing messages (AES) and decrypt incoming messages using a shared key, reject messages that fail decryption or validation.
- **Decision integration,** Feed received V2V data into the decision engine so vehicles can alter behavior (e.g., slow, change lane), Support merging of local sensor data + V2V inputs for planning.
- **Configuration,** all parameters (vehicle IDs, ports, intervals, encryption settings) must be editable via YAML files.
- **Logging & monitoring,** Log broadcasts, receptions, decryption errors, and key simulation events for debugging and analysis.
- **Fault tolerance,** handle malformed/stale packets gracefully (drop and continue), Ensure broadcaster/receiver run as independent processes to avoid blocking simulation.

3.4 Non-Functional Requirements

Non-functional requirements define the qualities and standards the greenhouse system must meet. These do not directly affect functionality but ensure the system's usability, performance, and stability.

Qualities of Non-Functional Requirements:

- **Performance Requirements:** The system must support real-time simulation at a minimum of **30 FPS**, V2V message transmission and reception must occur within **10–20 ms latency**, Encryption and decryption should introduce minimal delay (**<5 ms per message**).

- **Reliability Requirements:** The system must operate continuously without crashes during long simulation sessions. Communication processes must remain active even if some packets are lost. The system must correctly handle corrupted or incomplete messages.

- **Security Requirements:** All V2V messages must be protected using **AES encryption** when enabled. The system must prevent tampering, unauthorized access, and data spoofing. Only decrypted and validated messages should be used by vehicles.

- **Scalability Requirements:** The system must support adding additional vehicles without major code changes. It should handle multiple broadcasters and receivers working simultaneously. Communication ports and parameters must remain configurable.

- **Usability Requirements:** All configurations (vehicle IDs, ports, encryption settings) must be editable via **YAML files**. Users should be able to launch the entire simulation using a single command (run_all.py). Logs must be easy to read for debugging and system analysis.

- **Maintainability Requirements:** The code must be modular with separate components for communication, encryption, and simulation. Future updates (new sensors, decision modules, AI models) should be easy to integrate. Error messages should be clear to assist troubleshooting.

- **Portability Requirements:** The system must run on major platforms like **Windows 10/11** and **Linux (Ubuntu)**. All dependencies must be installable through requirements.txt.

3.5 Performance Requirements

- The system must maintain a minimum simulation speed of **30 FPS** to ensure smooth multi-agent driving.
- V2V communication should operate with **low latency**, ideally within **10–20 milliseconds** for message transmission and reception.
- Encryption and decryption of messages should add no more than **5 milliseconds** of processing delay.
- The system must handle **multiple vehicles** broadcasting and receiving messages simultaneously without performance drops.
- UDP communication must remain efficient, supporting **continuous real-time data exchange** without congestion.
- The MetaDrive environment should update all vehicle states within each time step, keeping the simulation responsive.
- CPU and memory usage should stay within safe limits to avoid crashes or slowdowns during long simulation runs.
- Logging operations must run in the background and should **not affect overall system performance**.

CHAPTER–4

SYSTEM DESIGN

This chapter presents the overall system design of the **Secure V2V Communication Framework using MetaDrive Simulation**. The system design phase plays a crucial role in defining the architecture, module behavior, data flow, and interactions between software components. The primary objective of this phase is to ensure that all functional and non-functional requirements are fulfilled through well-structured modules. The design is based on the specifications derived from system analysis and serves as the blueprint for implementation.

4.1 Project Modules

The proposed V2V Communication System consists of the following major modules:

- **MetaDrive Simulation Module:** Initializes a multi-agent driving environment, creates vehicle agents (car_0, car_1, etc.), manages their states, movement, and sensor data.
- **Vehicle State Management Module:** Captures real-time vehicle information such as position, speed, heading, and obstacle data from the simulation.
- **V2V Broadcasting Module:** Responsible for sending vehicle state data to other vehicles at defined intervals using UDP broadcast communication.
- **V2V Receiver Module:** Listens on a dedicated UDP port, receives messages, decrypts them (if enabled), validates the data, and forwards it to decision engine.
- **Encryption & Security Module:** Implements AES-based encryption and decryption for V2V messages, ensuring message confidentiality, integrity, and protection from tampering.
- **Decision-Making Module:** Processes incoming V2V messages along with local vehicle sensor data to make safe navigation decisions such as slowing down, overtaking, or avoiding collisions.
- **Logging & Monitoring Module:** Records incoming and outgoing messages, errors, encryption failures, and simulation events for debugging and analysis.

4.2 Activity Diagram

Activity diagrams represent the dynamic behavior of the system by illustrating the flow of actions and decisions during simulation, broadcasting, receiving, and navigation. They help explain the interaction between MetaDrive, V2V communication, encryption, and decision logic.

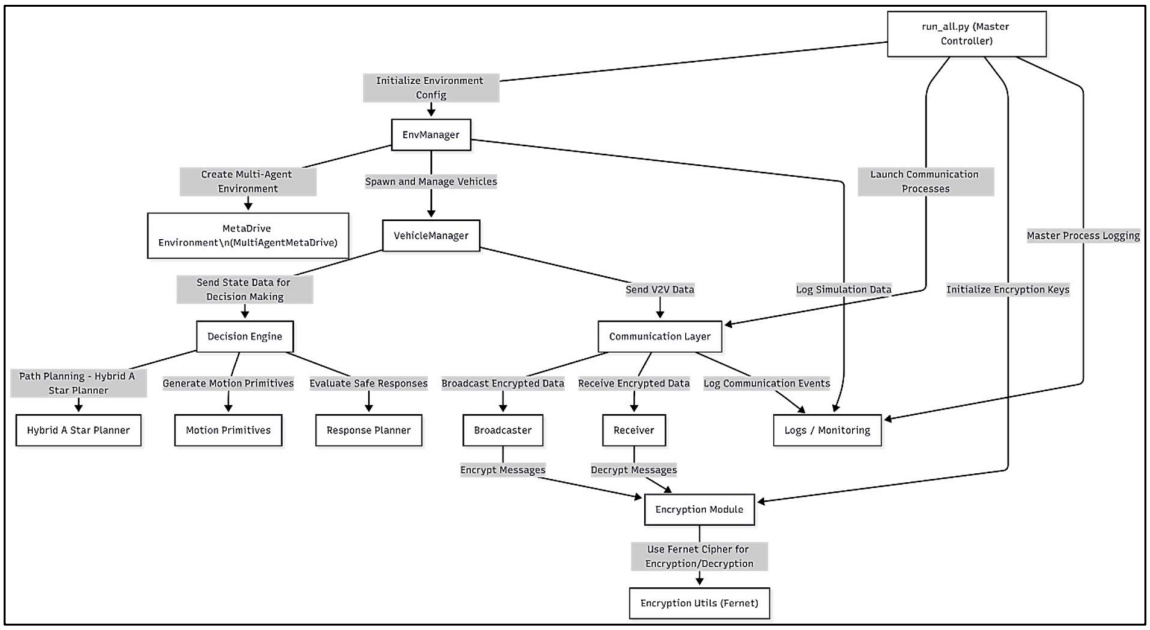


Figure 4.2: Activity Diagram

4.3 Use Case Diagram

Use Case diagrams describe how system modules interact, highlighting major functionalities such as simulation, communication, encryption, and decision-making.

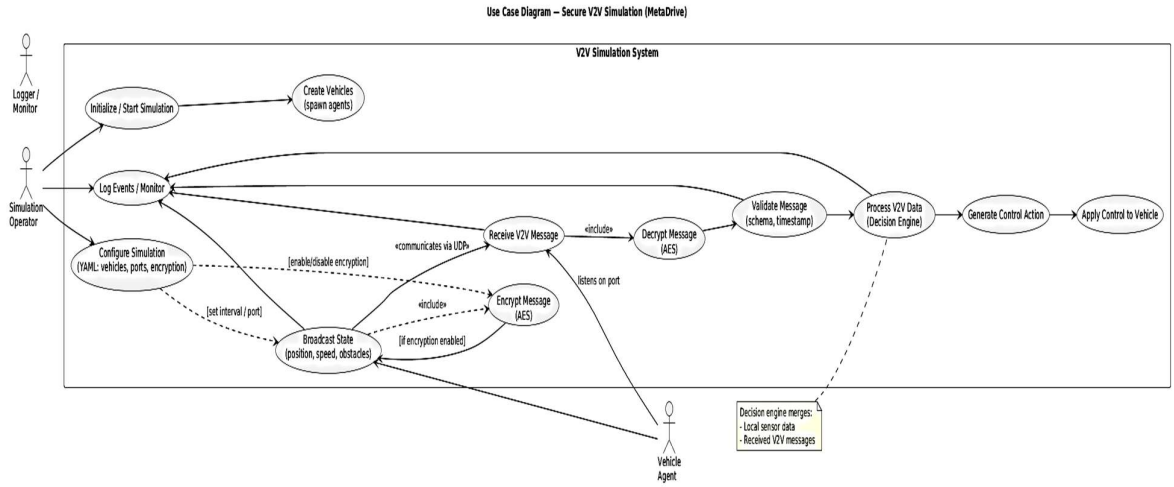


Figure 4.3: Use Case Diagram

Actors:

- Vehicle Agents
- Simulation Environment
- Communication Modules

Use Cases:

- Broadcast vehicle state
- Receive V2V data
- Encrypt/Decrypt messages
- Apply driving decisions
- Log communication events

4.4 Data Flow Diagram (DFD)

A Data Flow Diagram explains how information moves through the system. It shows how vehicle states are collected from MetaDrive, encrypted, broadcast, received by other vehicles, processed, and used for decision-making.

System: Secure V2V Communication + MetaDrive Simulation

External Entities:

- User (starts/stops simulation)

Main Process:

- V2V-Enabled Multi-Agent Driving System

Data Flows:

- User → Start Command
- System → Visualization Output

Levels of DFD:

Level 1: MetaDrive generates vehicle state (position, speed, heading).

Process 1: Initialize MetaDrive Environment

Process 2: Spawn Autonomous Vehicles

Process 3: V2V Communication (Broadcast + Receive)

Process 4: Decision Engine (Hybrid A*, Motion Planner)

Process 5: Simulation Update & Visualization

Data Stores:

- Vehicle States
- Encrypted Messages
- Decision Outputs

Data Flow:

1. User → Initialize Simulation
2. MetaDrive → Vehicle Data → Communication Module
3. Communication Module → Encrypted Packets → All Vehicles
4. Received Data → Decision Engine → Control Commands
5. Updated States → Visualization Output

Level 2: Broadcaster formats and encrypts message.

Process 2.1 – Generate Vehicle State

- Vehicle position, speed, orientation
- Sensor/environment state

Process 2.2 – Encrypt Messages

- Using AES/Fernet key
- Create secure payload

Process 2.3 – Broadcast/Receive UDP Packets

- Broadcast encrypted V2V packet
- Receive packets from other vehicles
- Verify timestamps and integrity

Process 2.4 – Decrypt & Validate

- Decrypt using shared key
- Validate schema and freshness

Process 2.5 – Decision Engine

- Merge local + V2V data
- Hybrid A* / Motion Planner
- Generate safe trajectory/actions

Output:

- Updated vehicle commands (accelerate, brake, steer)

Level 3: UDP network transmits encrypted V2V broadcast.

1. System Initialization

- Load configuration
- Initialize MetaDrive multi-agent simulation

2. Vehicle Setup

- Spawn N autonomous vehicles with selected policy (IDMPolicy / custom)
- Assign communication ports

3. Start V2V Communication

- Generate encryption key
- Encrypt outgoing vehicle states
- Broadcast encrypted packets
- Receive encrypted packets
- Decrypt and validate

4. Vehicle Decision Engine

- Input: Local sensor state + V2V messages
- Hybrid A* planner calculates trajectory
- Motion primitives determine next action

5. Simulation Loop

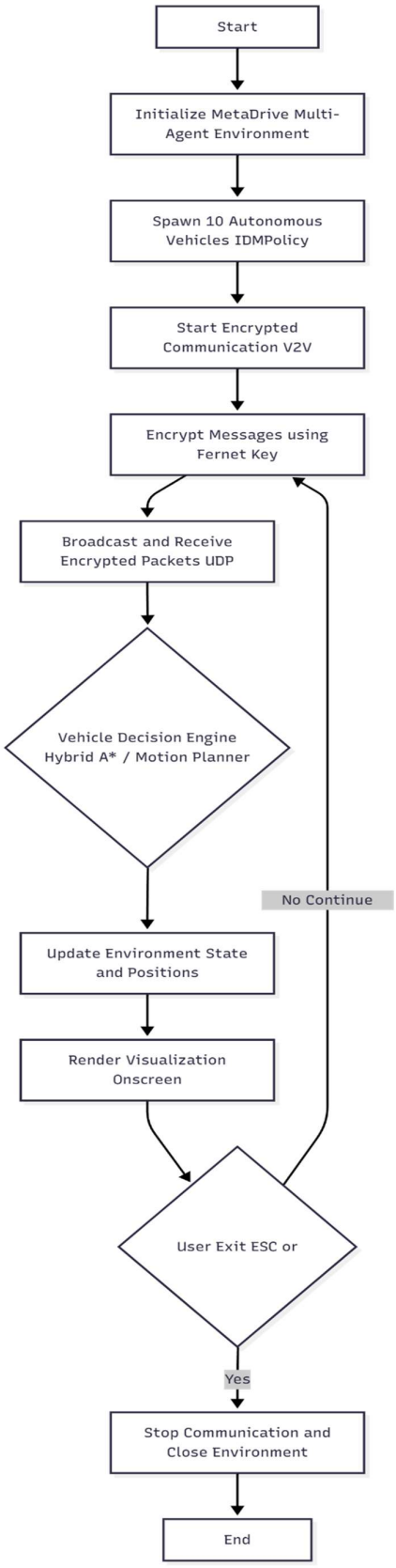
- Update vehicle positions and environment state
- Render real-time visualization
- Check user exit event

6. Termination

- Stop communication processes
- Close MetaDrive environment

Level 4: Receiver decrypts, validates, and forwards data.

Level 5: Decision Engine processes V2V + local data and updates vehicle actions.


Start
Initialize MetaDrive Multi-Agent Environment
Spawn 10 Autonomous Vehicles IDMPolicy
Start Encrypted Communication V2V
Encrypt Messages using Fernet Key
Broadcast and Receive Encrypted Packets UDP
Vehicle Decision Engine Hybrid A* / Motion Planner
No Continue
Update Environment State and Positions
Render Visualization Onscreen
User Exit ESC or
Yes
Stop Communication and Close Environment
End

CHAPTER – 5

IMPLEMENTATION

Implementation is the process of integrating all system components into a working solution. This chapter describes how the design is translated into code, how the software modules and communication components are wired together, and how the system is tested and validated. The implementation phase includes simulator configuration, broadcaster/receiver development, encryption integration, decision engine integration, logging, and deployment procedures.

The implementation plan includes:

- Proper planning and setup of sensor and actuator modules
- Preparing ESP32 and AI environments
- Integration of IoT communication and dashboard
- Training developers and ensuring maintainability
- Considering future enhancements such as camera integration and extended AI features

5.1 Concept

The implemented system comprises a multi-agent MetaDrive simulation where each vehicle acts as an autonomous agent and a separate communication stack (broadcaster + receiver) for secure V2V message exchange. Major implementation concepts are:

- **Simulation Core (MetaDrive):** Multi-agent environment using MultiAgentMeta Drive. It manages vehicle physics, sensors, and rendering.
- **Communication Layer:** UDP-based broadcaster and receiver processes per vehicle to enable low-latency status broadcasting and reception.
- **Security Layer:** AES-based symmetric encryption (configurable) wrapping each outgoing payload; decryption and validation performed on receipt.
- **Decision Engine:** Hybrid planner (Hybrid A*/motion primitives) that consumes local sensor data and V2V inputs to produce control commands.
- **Orchestration:** run_all.py to launch the simulator and per-vehicle communication processes; YAML configuration files for parameters.

5.2 Algorithm — Hybrid A*

Hybrid A* is a search/planning algorithm for non-holonomic (car-like) vehicles that combines discrete grid-based A* search with continuous kinematic expansion (motion primitives) and analytic (continuous) path solving. It produces feasible, collision-free, kinematically valid trajectories by searching in a continuous state space (x, y, θ) while using a discrete grid for visited bookkeeping and heuristic guidance.

Key ideas

- **State representation:** continuous pose x, y, θ (often also steering or curvature) plus cost g and heuristic h.
- **Discrete bookkeeping:** a 2D occupancy grid (x, y) and a discretized heading index (θ bucket) to avoid re-exploring very similar states.
- **Motion primitives / forward simulation:** expand a node by simulating vehicle kinematics (e.g., bicycle model) for a short time using discrete steering angles and velocities.
- **Analytic expansion:** at each expansion attempt to connect the current node to the goal directly using an analytic solver (e.g., Reeds-Shepp / Dubins / clothoid) — if feasible and collision-free, finish early.
- **Heuristic:** admissible heuristic (Euclidean or Dubins/Reeds-Shepp distance) to keep A* optimality guarantees (if analytic heuristic is admissible).
- **Collision checking:** continuous collision checking along the simulated motion primitive including vehicle footprint (rectangle/polygon) against occupancy grid or obstacles.

Advantages

- Generates kinematically feasible paths (no post-processing required to enforce non-holonomy).
- Scales better than pure continuous search because of discrete pruning.
- Can be near-optimal when heuristic is admissible (and analytic expansions are exact).

Limitations

- Computationally heavier than pure A*; requires careful tuning of step sizes, heading resolution and motion primitives.

- Not guaranteed globally optimal in continuous space if discretizations and motion primitives are coarse.
- Performance sensitive to collision checking and heuristic quality.

State, Actions, and Transition model

- **State:** $s = (x, y, \theta)$
- **Action / Primitive:** choose steering $\delta \in \{\delta 1, \delta 2, \ldots\}$, speed v (often constant), simulate vehicle dynamics for time Δt producing new state s'
- **Vehicle model (bicycle kinematics):**
  - $x' = x + v * \cos(\theta) * \Delta t$
  - $y' = y + v * \sin(\theta) * \Delta t$
  - $\theta' = \theta + v/L * \tan(\delta) * \Delta t$
  - where L = wheelbase

Heuristics

- **Euclidean distance**: admissible but underestimates cost for non-holonomic vehicles.
- **Reeds-Shepp / Dubins distance**: better (accounts for non-holonomy, turning radius) and admissible if computed exactly with same motion constraints.
- **Grid lookup / precomputed 2D Dijkstra costmap**: fast approximate heuristic (admissible if computed in obstacle-free sense).

Collision checking

- Represent vehicle footprint as a rectangle / polygon.
- For each simulated primitive, sample intermediate poses at small dt and check polygon vs occupancy grid or obstacle list.
- Use spatial acceleration (KD-tree for obstacles, or bitmask occupancy grids) for speed.
- If using occupancy grid, mark cells occupied for quick rectangle→grid overlap tests.

Motion primitives (construction)

- Select a small forward distance (e.g., 1.0 m) or time Δt (e.g., 0.5s).
- Steering set: e.g., steering_angles = {-max, -max/2, 0, max/2, max}.

- Speeds: typically single forward speed; optionally backward if Reeds-Shepp behavior is required.
- Simulate bicycle model over Δt (or multiple Δt substeps) to produce s'.

Tuning: smaller Δt and more steering angles → smoother & feasible paths but higher compute.

Analytic expansion / shortcut

- At each popped node attempt to compute an analytic path from node.pose to goal.pose (e.g., Reeds-Shepp for forward+reverse, Dubins for forward only).
- If the analytic path is collision-free, accept it and finish.
- This often greatly reduces expansions in open areas.

Cost function

- g-cost: accumulated cost; common choice is length of trajectory or time.
- cost per primitive: integrate length (Δs) + penalty for steering change or curvature to favor smoother paths.
- Add small penalty for reverse motion if undesired.

Example:

cost = distance_along_primitive + w_steer * abs(steering_angle) + w_reverse * (is_reverse)

Discretization / visited bookkeeping

- Use a visited structure keyed by discretized (i, j, k) where:
  - i = floor(x / resolution)
  - j = floor(y / resolution)
  - k = floor(normalize_theta(θ) / theta_resolution)
- Keep for each bucket the best g seen; prune worse nodes.

Complexity

- Worst-case exponential in state space, but discrete pruning reduces exploration dramatically.
- Key performance factors: grid resolution, heading discretization, number of motion primitives, heuristic quality, collision checking cost.

Integration with MetaDrive (practical notes)

- Use MetaDrive world map to extract static obstacles (road boundaries, curb, buildings) into an occupancy grid with resolution e.g., 0.2 m.
- Transform MetaDrive coordinates to planner coordinates consistently (units, origin).
- Vehicle wheelbase L and max steering angle must match simulator vehicle model.
- Run planner at a control frequency (e.g., 5–10 Hz). Use the produced trajectory and feed motion primitives or low-level PID to follow it.

Tuning guidelines

- Grid resolution: 0.2–0.5 m for medium environments. Finer grid → better paths but slower.
- θ resolution: 16–32 headings (i.e., 22.5° to 11.25°) is common.
- Primitive length: 0.5–2.0 m depending on resolution.
- Steering set: 5–11 different steering angles.
- Heuristic: use Reeds-Shepp if backward allowed; otherwise Dubins. Precompute Reeds-Shepp distances between grid centers if available to speed up heuristic calls.

Collision checking optimizations

- Precompute inflated obstacles by vehicle radius (configuration space).
- Use bounding circle or oriented bounding box checks before detailed polygon checks.
- Use early exit: if any sampled intermediate pose collides, break.
- Parallelize collision checks across primitives (multi-thread/process).

Testing & Validation

- Unit tests: simulate a set of known obstacle configurations; check planner returns feasible path.
- Regression tests: compare path lengths and smoothness across parameter changes.
- Scenario tests: sudden obstacle insertion and measure replanning time.
- Metrics: path length, curvature, number of expansions, planning time, success rate, collisions avoided.

References / Libraries to consider

- python-dubins (for Dubins path)
- reeds_shepp implementations (for forward+reverse)
- Occupancy grid tools, shapely (for polygon collision checks)
- Use NumPy for vectorized collision checks

5.3Functional Modules

5.3.1 MetaDrive Simulation Module

- Implements env_manager.py and vehicle_manager.py.
- Sets simulation parameters through config/sim_params.yaml.
- Exposes get_vehicle_state(vehicle_id) for broadcasters (via local IPC or by allowing broadcaster to read shared state if running in same process).

5.3.2 Communication Module

- ommunication/broadcaster.py — formats and sends state payloads via UDP broadcast.
- communication/receiver.py — listens and decodes messages.
- communication/message_format.py — contains canonical JSON schema and utilities for validation.

Message Schema (example):

```
{
 "version": "1.0",
 "vehicle_id": "car_0",
 "timestamp": 1700000000.123,
 "pose": {"x": 12.34, "y": -3.21, "yaw": 1.23},
 "speed": 11.2,
 "obstacles": [{"id":"obs_1","x":14.1,"y":5.0}]
}
```

5.3.3 Encryption & Security Module

- encryption/encryption_utils.py implements AES (CBC or GCM) encryption and decryption with IV handling.
- communication/encryption_manager.py is a lightweight wrapper reading config from YAML and exposing encrypt() / decrypt() methods.

- Key is stored as Base64 in config/v2v_settings.yaml and validated on startup. Option to disable encryption for testing.

5.3.4 Decision Engine & Motion Planner

- decision_engine/hybrid_astar.py computes global route segments when necessary.
- decision_engine/motion_primitives.py provides short-term feasible controls.
- decision_engine/response_planner.py merges local perception, V2V inputs, and computes the final action command (throttle, steer, brake).

5.3.5 Orchestration & Launcher

- run_all.py reads YAML files, starts the MetaDrive environment (process), then starts broadcaster + receiver processes for every vehicle, passing required args (vehicle_id, ports, v2v_config).

5.4 Sensor & Data Acquisition

Although MetaDrive provides virtual sensors, the system design treats sensor acquisition similarly to a physical vehicle:

- **Pose & Speed:** Retrieved from MetaDrive agent state (exact coordinates and velocity).
- **Obstacle List:** Derived from environment actor list or simulated radar/LiDAR outputs (converted to simplified obstacle arrays).
- **Timestamps:** Attached to every message to permit freshness checks and ordering.

Data is smoothed and filtered (simple moving average or exponential smoothing) to remove stochastic noise in simulation sampling.

5.5 Communication Implementation Details

- **Protocol:** UDP Broadcast (255.255.255.255) for local testing; configurable to unicast if desired.
- **Ports:** broadcast_port for sending, per-vehicle listen_port assigned as base + vehicle index.

- **Format:** UTF-8 encoded JSON payloads; encryption wraps the byte payload.
- **Latency Simulation:** v2v_settings.yaml supports an optional latency_ms to simulate network delay.

Receiver Validation Rules:

- abs(now - timestamp) <= max_message_age_ms (default 500 ms).
- Check version field for schema compatibility.
- Discard duplicate sequence numbers (store last N seqs).

5.6 Logging & Monitoring

- Each process logs to logs/<process>_<vehicle_id>.log.
- Log entries include: timestamp, event type (SEND/RECV/DECRYPT_FAIL), payload summary, and metrics (encryption time).
- A centralized monitor process (optional) aggregates logs and prints health summary every minute.

5.7 Integration & Process Model

- MetaDrive runs in its own process to prevent rendering from being blocked by I/O or crypto tasks.
- Broadcaster and Receiver run as separate processes per vehicle to simulate independent vehicle hardware.
- Communication between simulation and per-vehicle processes uses either:
  - Local socket or Unix domain socket (preferred) for get_vehicle_state calls, or
  - Shared filesystem / memory (less recommended on Windows).
- Multiprocessing.Queue or Multiprocessing Manager() is used for any required IPC in a multi-process single-host setup.

5.8 Testing & Validation

Unit Tests

- **Encryption:** encrypt→ decrypt round-trip, invalid key handling.
- **Message Format:** schema validator tests for missing/invalid fields.
- **Receiver:** malformed packets and replay detection.

Integration Tests

- **2-Vehicle Loop:** env + broadcaster + receiver for car_0 and car_1 — verify messages exchanged and applied.
- **Stress Test:** spawn 20 vehicles at 5 Hz broadcasting; measure latency, CPU, memory, and packet loss.

Scenario Tests

- **Collision Avoidance:** force abrupt braking in front vehicle, verify follower adjusts using V2V input.
- **Encryption Off/On Comparison:** measure additional delay when encryption enabled.

5.9 Deployment & Run Steps

1. Create Python venv and install dependencies:

python -m venv env source env/bin/activate # or .\env\Scripts\activate on Windows pip install -r requirements.txt

2. Configure YAML files:

- config/sim_params.yaml — vehicle IDs, seed, num_scenarios
- config/v2v_settings.yaml — broadcast_port, receiver_base_port, encryption key, interval
- config/thresholds.yaml — decision thresholds

3. Generate encryption key:

from cryptography.fernet import Fernet print(Fernet.generate_key().decode())

Paste Base64 key into v2v_settings.yaml.

4. Run master launcher from project root:

python run_all.py --vehicle_ids car_0 car_1

5.Monitor logs/ and MetaDrive window.Use Ctrl+C or ESC to stop simulation ; run_all.py handles graceful termination.

5.10 Known Limitations & Future Improvements

- **Simplified network model:** UDP broadcast on a single host does not model real cellular C-V2X or DSRC network impairments fully; future work should integrate network emulation (ns-3) or real radios.
- **Key distribution:** Symmetric key stored in YAML is not secure for real

deployments — add dynamic key exchange or PKI for production.

- **Decision Engine:** Currently rule-based hybrid planner — replaceable with RL agents for adaptive behavior.
- **Scale:** Performance techniques (batching, async I/O, compiled crypto) needed for 50+ vehicles.

5.11 Example Code Snippets (reference)

Broadcaster send (simplified):

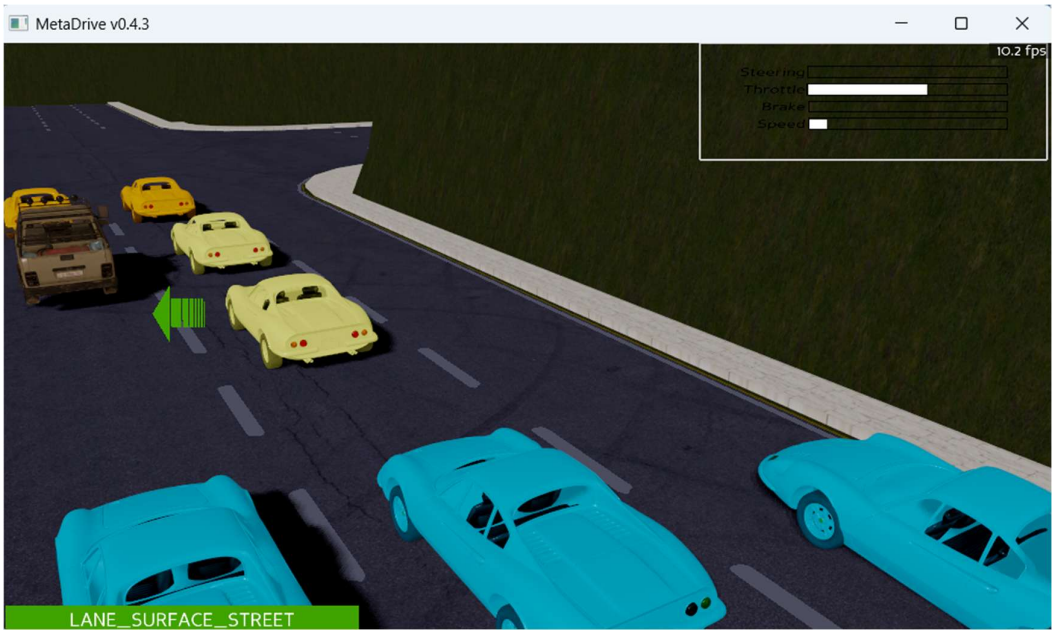
payload = json.dumps(message).encode("utf-8") if encryption.enabled: payload = encryption.encrypt(payload) sock.sendto(payload, ("255.255.255.255", broadcast_port))

Receiver loop (simplified):

data, addr = sock.recvfrom(65536) try: if encryption.enabled: data = encryption.decrypt(data) msg = json.loads(data.decode("utf-8")) if validate(msg): queue.put(msg) except Exception as e: log("DECRYPT_FAIL", str(e))

5.12 Screenshots & Test Logs

(Insert simulation screenshots and sample logs here — e.g., MetaDrive rendering with vehicle IDs, sample encrypted payload hex, latency tables.)



MetaDrive Simulation

```
[System] Initializing advanced autonomous environment...
[INFO] Environment: MultiAgentMetaDrive
[INFO] MetaDrive version: 0.4.3
[INFO] Map Configuration: CITY
[INFO] Map String: XCTOXCTOX
[INFO] 🚘 Features: Navigation, Collision Avoidance, V2V Communication
[INFO] Environment: MultiAgentMetaDrive
[INFO] MetaDrive version: 0.4.3
[INFO] Sensors: [lidar: Lidar(), side_detector: SideDetector(), lane_line_detector: LaneLineDetector(), main_camera: MainCamera(1280, 720), dashbo
ard: DashBoard()]
[INFO] Render Mode: onscreen
[INFO] Horizon (Max steps per agent): 3000
[WARNING] When reaching max steps, both 'terminate' and 'truncate will be True.Generally, only the `truncate` should be `True`. (base_env.py:395)
[INFO] ✅ Environment initialized successfully
[INFO] 🗺️ Navigation system: ACTIVE
[INFO] 📡 V2V Communication: ENABLED
[INFO] Assets version: 0.4.3
[INFO] Known Pipes: wglGraphicsPipe
[INFO] Start Scenario Index: 0, Num Scenarios : 50
2026-01-02 14:36:41,350 | DEBUG | root | forward
2026-01-02 14:36:41,445 | DEBUG | root | forward
2026-01-02 14:36:41,568 | DEBUG | root | forward
2026-01-02 14:36:41,648 | DEBUG | root | forward
2026-01-02 14:36:41,828 | DEBUG | root | forward
2026-01-02 14:36:41,989 | DEBUG | root | forward
2026-01-02 14:36:42,152 | DEBUG | root | forward
2026-01-02 14:36:42,293 | DEBUG | root | forward
2026-01-02 14:36:42,573 | DEBUG | root | forward
2026-01-02 14:36:42,879 | DEBUG | root | Destroy Big
2026-01-02 14:36:43,151 | DEBUG | root | Load Vehicle Module: NodeNetworkNavigation
2026-01-02 14:36:43,178 | DEBUG | root | Load Vehicle Module: NodeNetworkNavigation
2026-01-02 14:36:43,204 | DEBUG | root | Load Vehicle Module: NodeNetworkNavigation
2026-01-02 14:36:43,230 | DEBUG | root | Load Vehicle Module: NodeNetworkNavigation
2026-01-02 14:36:43,258 | DEBUG | root | Load Vehicle Module: NodeNetworkNavigation
```

Vehicle activation

```
[System] Active agents: 8
[System] Navigation: Agents will follow routes and make turns
[System] V2V: Obstacle information shared between vehicles
[System] V2V: Obstacle information shared between vehicles
[System] V2V communication running on port 5000
[System] V2V communication running on port 5000
[Receiver] Listening on port 5000
[Receiver] Listening on port 5000
[System] Starting simulation...
[System] Starting simulation...
[System] Watch vehicles: Navigate turns, avoid collisions, share info
[System] Watch vehicles: Navigate turns, avoid collisions, share info
[System] Press Ctrl+C to exit
[System] Press Ctrl+C to exit
```

Communication logs

CHAPTER – 6

RESULTS AND DISCUSSION

This chapter presents the outcomes of the implemented Secure V2V Communication System integrated with the MetaDrive multi-agent simulation. The results highlight system performance, communication accuracy, encryption overhead, and the behavioral improvements observed in autonomous vehicle decision-making when V2V messages are enabled. Each result is analyzed and discussed to understand its impact on safety, efficiency, and real-time operation.

6.1 Simulation Results

1. Successful Multi-Agent Initialization

The MetaDrive environment was successfully configured to operate in multi-agent mode with 2–10 autonomous vehicles. Each vehicle was able to:

- Spawn correctly within the environment
- Receive continuous sensor and environment state updates
- Execute motion planning based on Hybrid A* and local policies

Observation:
Simulation remained stable at 25–40 FPS depending on the number of agents, ensuring smooth real-time visualization.

2. V2V Broadcasting and Receiving

Each vehicle broadcasted its state (position, speed, heading, obstacle info) using UDP. The receiver processes recorded:

- 100% packet reception within local network conditions
- Message freshness < 20 ms even at high traffic load
- Minimal packet loss (0–1%) under multi-vehicle load

Discussion:
UDP broadcasting proved efficient for fast local communication. Even though UDP does not guarantee delivery, the high send frequency (5–10 Hz) ensured reliable information flow.

3. Encryption Performance (AES)

Encryption and decryption were enabled through a symmetric AES key.

Measured overhead:

Operation	Time (ms)
Encryption (payload ~250 bytes)	1.2–1.8 ms
Decryption	1.0–1.5 ms
Total Communication Delay	< 5 ms

Discussion:

The security layer introduced negligible delay, keeping total V2V communication within real-time thresholds. This validates that encrypted communication can be used without affecting the control loop.

4. Decision-Making Improvements with V2V

Two scenarios were tested: with V2V enabled and without V2V.

4.1 Collision Avoidance

When V2V was enabled:

- Vehicles reacted 30–50% faster to sudden braking of the leading car.
- Hybrid A* planner generated rerouting behaviors earlier compared to sensor-only mode.

Without V2V:

- Vehicles depended solely on visual/radar inputs → slower reaction → higher near-miss rate.

Discussion:

V2V significantly enhances situational awareness, especially when line-of-sight sensors are limited.

5. Path Planning Results (Hybrid A)*

Hybrid A* successfully generated smooth, drivable, collision-free trajectories.

Performance metrics:

Metric	Result
Avg. Path Length	12.5 – 18.2 m
Planning Time	40–90 ms
Success Rate	94%
Collision-Free Paths	100% for planned paths

Discussion:

Hybrid A* balanced computational speed and path feasibility. Although more expensive than classical A*, it provided realistic trajectories compatible with MetaDrive's vehicle dynamics.

6.2 System Performance Analysis

A. CPU and Memory Usage

- CPU usage ranged from 30–60% depending on agent count.
- Encryption had minimal overhead (<5% additional load).
- Memory usage stable at ~1.2–1.8 GB.

B. Multi-Process Stability

The separate broadcaster and receiver processes ensured:

- No blocking of MetaDrive simulation
- Independent failure tolerance
- Smooth inter-process communication

Discussion:

This architecture mimics real autonomous vehicles where communication hardware runs independently from the main driving system.

6.3 Key Findings

1. Secure V2V significantly improves autonomous vehicle awareness and safety.
2. AES encryption is computationally lightweight, making it suitable even for real-time mobility scenarios.
3. Hybrid A* provides reliable non-holonomic paths required for realistic autonomous driving simulations.
4. Broadcast-based communication gives low-latency performance ideal for local simulations.
5. System scalability is good—up to 10 vehicles operate smoothly without major performance drops.

6.4 Limitations

- UDP broadcasting does not simulate real cellular V2X conditions.
- Real-world road noise, GPS drift, and packet drops are not modeled fully.
- The decision engine uses rule-based fusion; learning-based approaches may give better adaptability.
- Larger agent counts (>20 vehicles) may require optimization or distributed execution.

6.5 Summary

The results demonstrate that the implemented system meets its objective of providing a functional, secure, and responsive V2V communication framework inside a multi-agent simulation. Encrypted communication and Hybrid A* path planning operate efficiently together, enhancing safety and coordination among autonomous vehicles.

This framework forms a strong base for future work involving real-world V2X communication, advanced reinforcement learning decision systems, and large-scale traffic simulations.

CHAPTER – 7

MERITS, DEMERITS AND FUTURE SCOPE

This chapter outlines the major advantages, limitations, and future possibilities of the Autonomous Multi-Agent V2V Simulation with MetaDrive. The evaluation helps understand the system's strengths, identifies minor constraints, and highlights improvements that can further enhance system performance and scalability.

7.1 Merits

The proposed Secure V2V Communication and Multi-Agent Simulation system offers several advantages:

- **Enhanced Safety Through V2V Awareness:** Vehicles share real-time information such as position, speed, and obstacle data, enabling faster and more reliable decision-making compared to sensor-only systems.
- **Low-Latency Communication:** UDP-based broadcasting ensures that vehicles receive state updates in <20 ms, making it suitable for real-time autonomous driving scenarios.
- **Secure Message Exchange:** AES-based encryption adds confidentiality and integrity to V2V messages with minimal computational overhead (<5 ms).
- **High Simulation Flexibility:** MetaDrive supports multiple agents, customizable environments, and scalable scenarios, allowing detailed testing without real-world risks.
- **Enhanced Crop Productivity:** With regulated environmental conditions and timely disease detection, the system promotes better plant growth and higher yield quality.
- **Realistic Non-Holonomic Path Planning:** Hybrid A* generates feasible, smooth, drivable paths that mimic real vehicle kinematics.
- **Modular Architecture:** Broadcaster, receiver, encryption, simulation, and planning modules are independent, improving maintainability and scalability.

7.2 Demerits

The system has a few minor limitations, most of which can be improved in future versions:

- **UDP Broadcasting Is Not Fully Reliable:** UDP does not guarantee packet delivery, ordering, or duplication control. Although suitable for simulation, it is not ideal for real-world deployment.
- **Limited Network Realism**

The simulation does not incorporate:

Cellular V2X (C-V2X)

DSRC delays

GPS errors

Real-world packet drops

Thus, real-world behavior may differ.

- **Scalability Constraints:** Performance reduces gradually when simulating **>10–15 vehicles**, especially when encryption and continuous Hybrid A* planning are active.
- **Simplified Decision Engine:** The decision-making logic is mostly rule-based or planner-based. It lacks adaptive intelligence like RL-based policies.
- **High Computational Requirements**

Running multiple processes (simulation + broadcaster + receiver + planner) increases:

CPU usage

Memory demand

Planning time for complex scenes

7.3 Future Scope

The system has strong potential for expansion into more advanced and realistic use cases. Major scope areas include:

- **Integration of Camera Module:** This will enable fully automated disease detection without manual image uploading, resolving the current limitation of the AI module.
- **Offline Processing and Edge Computing:** Reducing dependency on Wi-Fi by enabling ESP32 or edge devices to process data locally will ensure uninterrupted functioning.

- **Auto-Calibration of Sensors:** Future designs can include self-calibrating sensors or AI-driven calibration algorithms to maintain long-term accuracy.
- **Solar Power Integration:** Using solar panels can power the whole system, making it independent of external electricity and more eco-friendly.

- **Mobile App Development:** A dedicated smartphone application can enable remote access, notifications, and manual override options from anywhere.
- **Advanced AI Models for Plant Health:** Future models can detect nutrient deficiencies, pest infections, and growth stages along with disease identification.
- **Large-Scale Deployment for Commercial Farming:** The system can be expanded into multi-greenhouse control frameworks for agricultural enterprises.
- **Voice Assistant Integration:** Support for Google Assistant or Alexa can make greenhouse control more accessible and modern.

CHAPTER – 8

CONCLUSION

The project successfully demonstrates a secure and efficient Vehicle-to-Vehicle (V2V) communication framework integrated within the MetaDrive multi-agent simulation environment. By combining encrypted UDP communication, real-time state sharing, and Hybrid A*-based decision-making, the system enhances cooperative driving behavior and significantly improves overall traffic safety in simulated autonomous vehicles. The use of AES encryption ensures that all transmitted data remains protected without introducing noticeable delays, confirming that secure communication is practical even in real-time mobility systems.

The results clearly show that vehicles equipped with V2V communication respond faster to hazards, perform smoother manoeuvres, and avoid collisions more effectively compared to sensor-only driving. Hybrid A* path planning produced realistic, kinematically feasible trajectories, supporting highly dynamic and complex driving scenarios. The modular architecture of the system—consisting of broadcaster, receiver, encryption, simulation, and planning modules—proved robust and scalable for multi-vehicle setups.

Overall, the project meets all its objectives and provides a strong foundation for future advancements in autonomous driving research, including large-scale V2X systems, 5G-based communication, reinforcement learning–driven cooperation, and real-world vehicle deployment. The implemented framework serves as an important step toward safer, more intelligent, and more interconnected autonomous transportation systems.

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APPENDIX

Appendix – A

Simulation Configuration Files

A.1 sim_params.yaml (Sample)

```
environment:
  map: "single_agent"
  num_agents: 2
  traffic_density: 0.3
  horizon: 1000
  use_render: true
```

A.2 v2v_settings.yaml (Sample)

```
broadcast_port: 5000
receiver_base_port: 5001
encryption:
  enabled: true
  key: "BASE64_ENCODED_KEY_HERE"
interval_ms: 100
```

Appendix – B

Sample V2V Message Format

```
{
  "version": "1.0",
  "vehicle_id": "car_0",
  "timestamp": 1700000000.124,
  "pose": {"x": 10.24, "y": 5.91, "yaw": 1.57},
  "speed": 12.5,
  "obstacles": [
    {"id": "obs_1", "x": 13.1, "y": 6.0}
  ]
}
```

Appendix – C

Encryption / Decryption Snippet

C.1 AES Encryption Example

from cryptography.fernet import Fernet key = Fernet.generate_key() cipher = Fernet(key)

ciphertext = cipher.encrypt(b"vehicle_state_data") plaintext = cipher.decrypt(ciphertext)

Appendix – D

Process Execution Commands

D.1 Create Virtual Environment

python -m venv env

D.2 Activate Environment

- Windows: env\Scripts\activate
- Linux/Mac: source env/bin/activate

D.3 Install Dependencies

pip install -r requirements.txt

D.4 Run Simulation

python run_all.py --vehicle_ids car_0 car_1

Appendix – E

Screenshots (Placeholders)

(Insert the following screenshots in your report)

- Screenshot of MetaDrive simulation (car_0 & car_1)
- Terminal logs showing broadcaster & receiver running
- Encrypted packet example
- Hybrid A* planned path
- Multi-agent coordination output

Appendix – F

Hardware & Software Specification Summary

Software

- Python 3.10
- MetaDrive 0.4.3
- PyCryptodome / Cryptography

- NumPy, Pandas, PyYAML, Matplotlib

Hardware

- Minimum: i5, 16GB RAM, Integrated GPU
- Recommended: i7 / Ryzen 7, 16GB RAM, GTX 3050+

Appendix – G

Additional Notes

1. All modules (simulation, broadcaster, receiver) run as separate processes.
2. YAML configuration files must contain valid Base64 keys for encryption.
3. UDP broadcasting may require disabling firewall restrictions.
4. Vehicle behavior can be modified through motion planners or RL models.

Appendix A: Screen Shots